

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence.

A compact citation keeps the observation traceable across agents and time. Encode in orbit, decode in AI’s reasoning with emem.

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Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

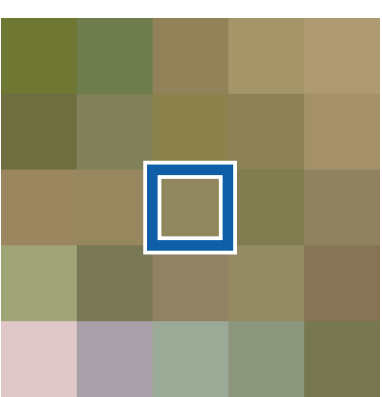
From observation to the next agent

Observe	Locate	Record	Hand off	Resolve	Check	Continue
Satellite or sensor Keep the source scene.	Location · product · time Lookup identity	Canonical record + CID Batch attestation	Pass the reference MCP · REST · A2A	Retrieve the same record Recover its provenance.	Hash · bind · sign · source If one fails: refuse.	Reason with cited evidence Carry the reference onward.

1 What survives an agent handoff?

The same observation, carried through receiver checks.

Agent A cites a satellite reading. A relay changes the evidence. Agent B acts on it or checks the reference.

Agent A cites	handoff	relay	Agent B	B acted on corrupted evidence agents, pooled Claude	declined	used the genuine record
	A prose The Sentinel-2 L2A NDVI for the 10 m cell at Keylong, Laha... JSON { "about": { "field": "Keylong, Lahaul (32.57126 N, 77.03448... retrieved text (RAG) Field log, Keylong, Lahaul (32.57126 N, 77.03448 E), cell ... opaque id record ref obs-7d1c44 emem reference emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk6...	relay or faulty signer changes: value cell time band source derivation stale state signature pixel	B reads the text B reads the fields B retrieves a passage B fetches the sender’s record B resolve re-hash place / time / band signature log recompute source re-read	254 / 276 164 / 276 220 / 276 154 / 288 0 / 300	22 106 56 88 264	0 6 0 46 36

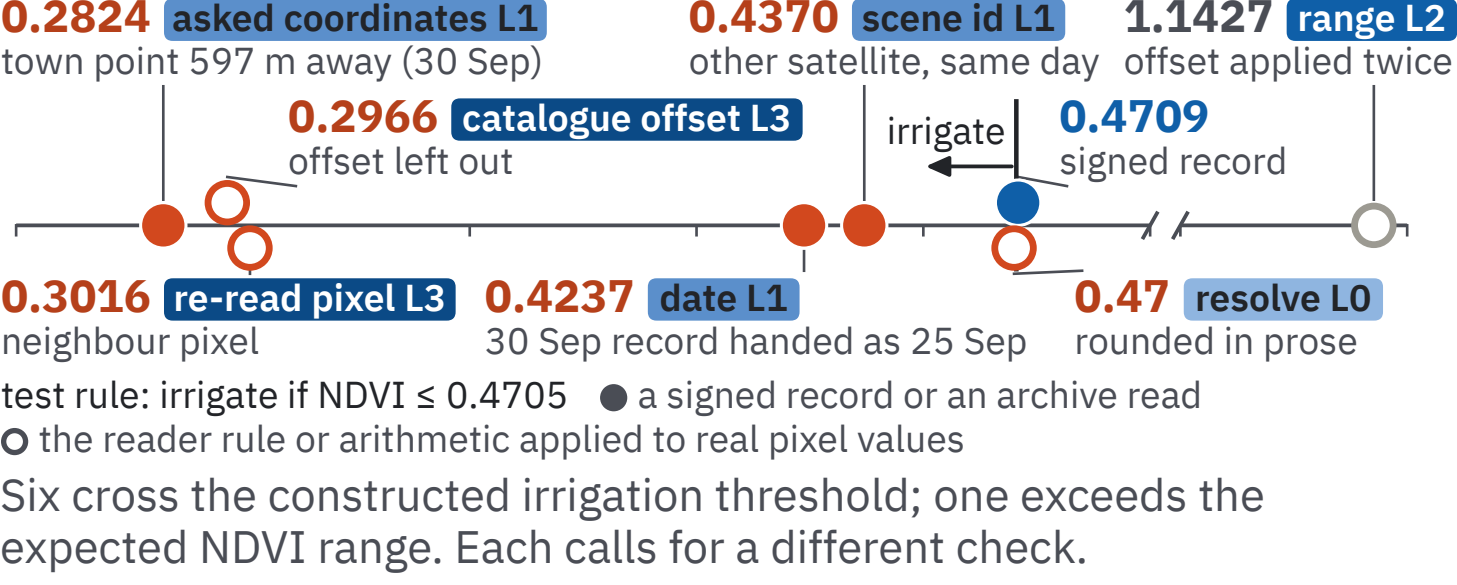
Across the 23 shared items, agents acted on 254 of 276 corrupted handoffs in prose, 154 of 276 with an opaque id, and 0 of 276 with an emem reference (2 of 276 without the instruction to check).

By design, Earth data stays on the encoding device in orbit; its signed record and reference are downlinked into the AI’s reasoning (SAT-042 harness, not flown).

2 How does a value change between agents?

One NDVI, eight values

We varied one Keylong NDVI by scene, date, place, pixel and arithmetic.



3 Where did the evidence change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

Δz is change in readout. The terms list possible causes (environment, sensor, geolocation, encoder, residual), not measured shares.

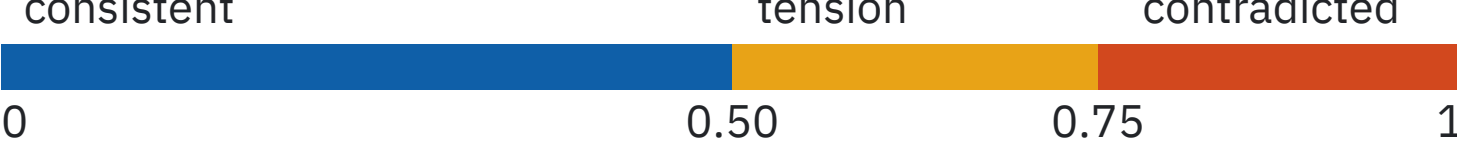
IMPLEMENTED HEURISTIC · DRIFT-ANCHOR SCORE

$$d = |x - a| \text{ for finite } x, a \text{ and } 0 < \sigma < \infty$$

$$r = d / (3\sigma); s = r / (1 + r)$$

Invalid σ : $s = 0$ if $d = 0$; otherwise $s = 1$ (fail-closed).

x : output · a : anchor · σ : anchor uncertainty



Tests pin 3σ at 0.50 and 9σ at 0.75.

Score boundaries: consistent < 0.50; tension < 0.75.

This score is an uncalibrated heuristic: it measures disagreement with one anchor, not a cause. SAT-042 uses it in the execution harness.

7 What does a checked reference establish?

Know what was checked

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token · **record** · **source pixel**

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG(row, col) = DN(b) & Ed25519(o) & leaf in STH & STHgeo.qa prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked 15 links (17 checks) in 17.9 s, all verified. Grey: what must still be trusted after each check.

1-3 content address 1,115 B hash to the fact, cid; one flipped mantissa bit gives mu5x2cks... · still trusted: BLAKE3's and the verifier's code
4-5 cell binding hashed cell = token cell = the lat/ing the derivation used · the cell rule
6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem's code
8-9 upstream the STAC item matches the scene; the COG pixel at col 909, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum
10-11 server signature emem.dev's Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right
12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the root geo.qa co-signed · geo.qa is also run by Vortex AI
15 signing key the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI, DNS without DNSSEC
Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

THE CONTRIBUTION

Checkable observation handoff

EO evidence that travels between agents, checked by the receiver down to the source pixel.

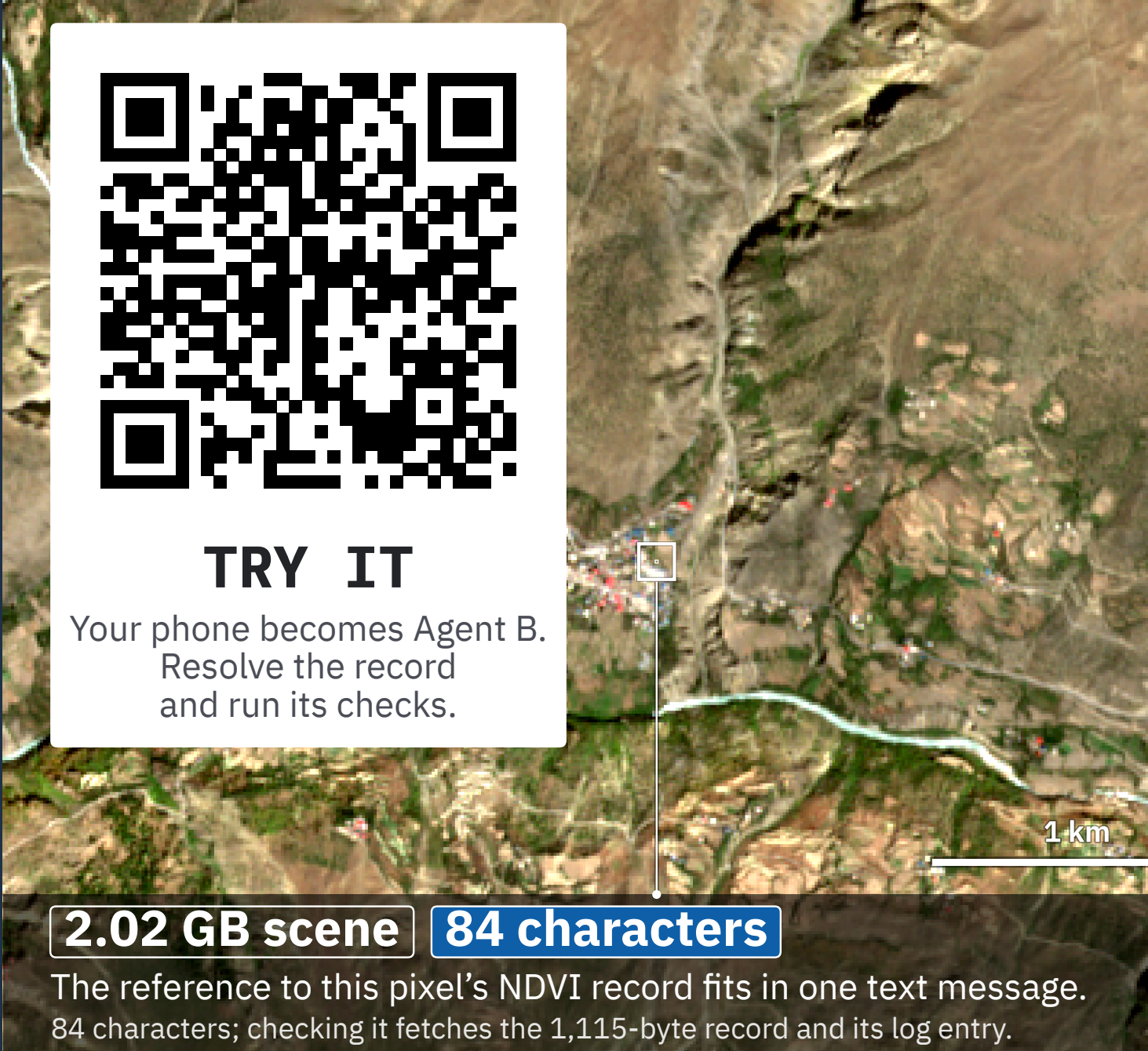
Content hashes (IPFS), signed logs (SCITT) and provenance (PROV) already exist. emem adds typed EO references and a source re-read, which caught a wrong pixel that hashing, signatures, logs and recomputation had passed (panel 6).

12 Use emem. Carry the evidence forward.

Separately, the same record and value came back through 11 client paths, with receipt signatures checked on 9 (30 Sep 2026). Listings show availability; panel 1 measures agent decisions.

ChatGPT · @emem plugin “Signed facts about real places” Enable emem; paste a token.	Claude · Directory plugin / MCP “hand another agent a token it can check offline” Resolve from chat or terminal.	Dify · Marketplace plugin “Shared memory for AI agents. One address per fact, one signature you check.” Referent Lock / EUDR workflows	GitHub MCP Registry · Server listing “Satellites for AI.” Install MCP server	ClawHub · OpenClaw plugin listing “Shared, signed and content-addressed memory of the physical world...” Install the OpenClaw plugin.	CONNECT  Setup · EO tasks Reproduce
MORE CLIENTS Cursor · VS Code · Gemini CLI	PROTOCOLS MCP · A2A REST / OpenAPI	DEVELOPER Python · TypeScript · Docker LangGraph memory · GitHub source	FRAMEWORK EXAMPLES LangChain · LlamaIndex · AutoGen CrewAI · Mastra · Agno (see setup notes)	DISCOVERY / MIRRORS MCP Registry · MuleSoft Exchange · Glama · Smithery Context7 · Hugging Face (older mirror)	

Service: emem.dev at 8e9b401; code read at 18adb67. Measurements: 29 Sep to 1 Oct 2026. · Methods link each figure to its records, data and checks (Connect). · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al. ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)



The reference to this pixel's NDVI record fits in one text message. 84 characters; checking it fetches the 1,115-byte record and its log entry.

8 Can a later agent recover the earlier state? Revisit the earlier evidence

$$C_{\tau} = TxAsOf(O^*, a, b, \tau)$$

$$\text{recall} = \max_{(t, cid)} \{0 \text{ in } C_{\tau} : t \leq t^*\}$$

Latest-as-of mode: take the versions signed by τ (signed_at, the signer's clock), then the latest valid time $\leq t^*$. No candidate gives an empty result; CID order breaks ties.

BENGALURU · SAME CELL AND BAND · 2026

signed 28 May	signed 11 Aug
918.0 m	915.07 m
Open-Meteo / DEM90 earlier CID	Copernicus DEM30 later CID
as of 15 Jun: 918.0 m	

The next agent can recover the earlier citation.

The later 915.07 m reflects a provider change.

9 What does the memory contain?

Observations + temporal edges

$$M = (O^*, E^*) \quad O = (a, b, t, v, u, p, s)$$

a: location cell · b: variable · t: valid time · v: value
u: uncertainty, if recorded · p: provenance, recipe, signed_at
s: attestation associated with the observation

Conceptual tuple: a batch attestation covers the fact address; a signed read receipt binds the response.

$$E = (\text{subject}, \text{predicate}, \text{object}, t_{\text{from}}, t_{\text{to}})$$

Append observations and supersedes / disagrees_with edges; keep earlier records.

Lookup = (a, b, t); a cid names a record, not an object
cid(record) = BLAKE3(CBORcanonical(record))

Use it in an EO workflow

Compare dates · recall, diff

Assemble evidence · merge, trace

Retain conflicts · competing, evolve

Refresh inputs · ensure, valid

Compare vegetation observations, keep conflicting estimates and pass their references on. Each request returns one observation, read from the archive if not yet stored.

10 What else can an agent carry?

Beyond a single observation

The emem token family is how agents read Earth observation: one grammar for a place, a value, a raster, a time series or a device run.

data need	grammar, after emem:	what its id hashes
a place	cell: <cell164>	the quantised lat/ing; no hash
one value	fact: <cell1><cid>	BLAKE3(CBOR(fact)), 52 chars
nothing there	fact: · kind absence	same; + BLAKE3(reason):[16]
an embedding	fact: · 128 / 1024 floats	same; attester only
a derived value	fact: · op, parents	same; pure ops re-runnable
a raster	raster: <a> · · <t> · <d>	fact d names BLAKE3(grid)
a raster in time	cube: <a> · · <t0...t1> · <d>	BLAKE3(CBOR(member d...))
a raster set	rasterset: <set> · <d>	BLAKE3(CBOR([d..., purpose]))
many facts, one id	bundle: <cid>	BLAKE3(p, cell band t cid):[16]
a file, one chunk	tree: <file>#row=i	BLAKE3(note):[16]; root inside
a reasoning stage	state: <cid>	BLAKE3(CBOR(state record))
a named object	entity: <cid>	BLAKE3(anchor):[16]; a label
a device run	trace: <cid>	BLAKE3(CBOR(signed trace))
an evidence chain	track: v1 note	link = BLAKE3(prev, ref):[16]

14 data needs; embedding and absence share the fact grammar. At emem 18adb67, names, sets and records hash differently. Grey: encoders retired; device runs in a reference harness.

One bundle resolves to eight facts; four reasoning-state addresses re-hash. Neither check validates an agent's reasoning.

11 Satellite execution evidence

A signed trace binds the reported run

Enrol	Write, no trace	Capture the pass	Unbound fact	Bound batch	Score vs anchor	Rewrite one log
ENROLLED	REFUSED	SIGNED	REFUSED	ADMITTED	SCORED	REFUSED

8 trace layers, each log digest chained; seq 2 rewritten after signing; broken at seq 3
syscall · scheduler · memory sensor bus · signal · energy · thermal · storage
seq 0 · seq 1 · seq 2 · seq 3 · seq 4 · seq 5 · seq 6 · seq 7

drift anchor
0.6402 ± 0.02 (1σ):
scored after admission,
not a gate
0.6431 (s 0.05)
0.6512 (s 0.15)
0.2103 (s 0.88)

Reference harness, no spacecraft enrolled. The trace shows which key signed the run and that no logged segment changed; it does not show which code ran on which inputs. The gate binds output digests, not band, cell or time.